

Psychon Phase Mechanics

The Interior State of a Pruned Agent, the Inaccessibility of Its Path,
and the Gap-Direction Definition of a Question

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Abstract

The companion specification constructs an individual by pruning a thought-pattern module against a demographic substrate, and stops at the resulting finite weighted graph. This paper takes that graph as given and asks three questions the construction leaves open: what state does an agent carry on it, how much of that state is available to any other agent, and what is the complete set of things one agent can do to another.

We answer with a single object. A *psychon* is a pair $\Psi = (\gamma, v)$ of a walk through the agent's graph and the vertex it ends at, subject to a monotone step count. We prove, in order: (i) a *separation theorem*, that every vertex of a finite weighted graph with a distinguished medium vertex has a well-defined *resting cut*—the minimum-weight boundary separating it from the rest—computable by max-flow, whose cost is bounded below by the floor $\beta > 0$ inherited from the parent construction, so a psychon terminates somewhere definite and never on a boundary of zero thickness; (ii) a *recognition-search identity*, that under a monotone record no operation on the graph has the signature of retrieval, so recalling is searching and the distinction between memory and inference is not expressible in the runtime; (iii) a *path-opacity theorem*, that distinct trajectories reach identical termini with identical step counts, so the pair (γ, v) is strictly finer than anything an agent can emit, and the emitted state is exactly the reduced pair (v, m) of terminus and count; (iv) a *receiver-relativity theorem*, that one event registers a different resting cut in every agent that receives it, that each such registration is correct on its own graph, and that divergence between receivers is therefore not error and admits no adjudication; (v) a *drift theorem*, that composed re-registrations across a chain of receivers have no author, since the composite differs from every individual contribution and no contribution is distinguished; (vi) an *act-direction theorem*, that acts partition exhaustively into two kinds by their effect on the residual gap—a *report* lowers the gap on the acting side, a *question* raises it on the receiving side—and that this is the only non-vacuous distinction available, since neither content nor intent is expressible; (vii) an *adoption theorem*, that two agents of unequal internal structure attain equal behaviour when their boundary conditions match, on a timescale set by coupling above a critical threshold, so that a practice propagates without any interior being shared or inspected.

We close by fixing the correspondence between the discrete pair (v, m) and a continuous gradient flow on a bounded coordinate box, showing the two descriptions agree on the attracting set, and stating precisely which quantities of the continuous theory are *not* observables of the instrument. Nine implementation invariants follow, each a decidable predicate on the runtime.

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1 Introduction

1.1 What the parent construction leaves open

The companion specification builds a population as a family of *modules*—coarse regions of a shared substrate graph, each a thought-pattern held by many—and produces an individual by *pruning*: intersecting the modules a demographic record admits, and taking the subgraph induced on what survives. The construction is deterministic before contact, and the resulting object is a finite weighted graph.

It stops there, and it must, because at that point the agent has no history. Two agents pruned from identical records are identical graphs; the parent manuscript proves they are not yet individuals. What makes them individuals is what happens on the graph afterwards. That is the subject here.

Three questions are open, and they are not independent.

- (a) **State.** An agent is somewhere on its graph and got there somehow. Is the “somehow” part of its state? If it is, the state is a walk and grows without bound. If it is not, two agents with different histories are the same agent, and the individuation claim of the parent manuscript fails.
- (b) **Availability.** Whatever the state is, some function of it is emitted when the agent acts. If the emitted function were injective, an observer could reconstruct the interior, and the non-attribution results of the parent manuscript would be false. If it were constant, agents could not be told apart at all.
- (c) **Acts.** The runtime has to enumerate what one agent can do to another. Content-based taxonomies (assert, ask, promise, deny) are unavailable: the parent manuscript proves the runtime cannot evaluate the truth of a content, and an agent cannot certify its own interior, so any classification appealing to what is said or meant is not a predicate the implementation can decide.

We answer all three with one object, and the answers turn out to be forced rather than chosen.

1.2 The object

A *psychon* is a pair

$$\Psi = (\gamma, v), \quad (1)$$

where γ is a walk on the agent graph and $v = \gamma(k)$ is the vertex it currently occupies, carried together with a monotone count m of committed steps. The pair is the state of the agent. The count is what it emits. The walk is what nobody, including the agent, can report.

This resolves (a) and (b) simultaneously, and in the only way consistent with the parent axioms. The walk is genuinely part of the state—so two agents that reached the same vertex by different routes are different agents, which is what individuation requires—while the emitted quantity is the reduced pair (v, m) , which is strictly coarser, so reconstruction of the interior is impossible rather than merely difficult. Question (c) is answered in [Section 7](#): the only distinction between acts that survives the loss of content is the *direction* in which an act moves the residual gap, and that distinction is exhaustive.

1.3 Relation to the parent manuscript, and to prior frameworks

This paper is a companion, not a revision. It imports the floor, the agent construction, the attention allocator, and the coordination threshold, and adds no axiom that the parent does not already state or immediately imply. Where the parent proves that the instrument *cannot* report a verdict or an attribution, this paper supplies the interior mechanics that make those impossibility results properties of the object rather than restrictions on the interface.

A remark on scope is owed to the reader, because several results below have recognisable analogues in continuous and physical settings: order parameters, gradient flows, synchronisation thresholds. We use those analogues where they are load-bearing and name them where they are not. Specifically, we adopt a gradient-flow description of the attracting set ([Section 9](#)) because it supplies an explicit potential for an attractor that the parent manuscript proves exists but does not construct; and we adopt a coupling-threshold form of the adoption result ([Section 8](#)) because propagation without shared interiors is exactly the phenomenon the instrument is built to study. We do *not* adopt any global order parameter as an observable, and [Section 9.1](#) states precisely why: a quantity defined over the whole population is a world-state read-out, and [Theorem 6.2](#) forbids one. Every continuous object in this paper is defined on the pruned graph of a single agent and is receiver-relative by construction. Readers approaching from the physical literature should treat that restriction as the substantive claim of [Section 9](#), not as a limitation of it.

2 Primitives and axioms

Definition 2.1 (Agent graph). An *agent graph* is a finite, connected, weighted, undirected graph $G = (V, E, w)$ with $w : E \rightarrow \mathbb{R}_{>0}$, together with a distinguished vertex $\perp \in V$ adjacent to every other vertex, called the *medium*. We write $n = |V| \geq 2$.

The medium is not a thought. It is the vertex standing for everything the agent has not distinguished: the remainder of the world as it bears on this graph. Its role is to make “separating something from the rest” a well-posed finite question. In the runtime it is the vertex produced by pruning that carries no module of its own.

Axiom 1 (Distinction floor). There is $\beta > 0$ with $w(e) \geq \beta$ for every $e \in E$.

Axiom 2 (Search). Every operation by which the agent arrives at a vertex is a walk along edges of G from the vertex it currently occupies. There is no operation whose effect is to place the agent at a vertex not adjacent to its current one.

Axiom 3 (Act floor). There is $\varphi > 0$ such that every committed step of a walk increments the record m by at least φ . The record is monotone: no operation decrements it.

Axiom 4 (Non-completion). No walk exhausts G : for every walk γ of finite length there is a vertex not on γ and an edge not traversed by γ .

[Axiom 1](#) is the floor of the parent manuscript, restricted to one pruned graph. [Axiom 2](#) is its locality of conversation. [Axiom 3](#) is what makes the record a record: an act costing nothing would leave no trace and could be performed unboundedly often between two observations. [Axiom 4](#) holds in the runtime by construction, since the substrate is strictly larger than any pruned subgraph and the medium is always available as an unexhausted neighbour.

Remark 2.2 (The floor is not a modelling convenience). A boundary of weight zero is not a boundary: if some e had $w(e) = 0$, its endpoints would be separable at no cost and would not be two things. So [Axiom 1](#) is constitutive of there being a graph with more than one vertex at all, rather than an assumption added for tractability. Note also that on a finite graph the floor is attained automatically, since $\beta = \min_{e \in E} w(e)$ exists and is positive; we state [Axiom 1](#) separately only because the constant β appears in the bounds below.

3 The resting cut

Before we can say what a psychon terminates *at*, we need the sense in which a vertex is a definite thing on its graph. That is a separation question, and it has an exact finite answer.

Definition 3.1 (Separation cost and resting cut). For $v \in V \setminus \{\perp\}$, let

$$\sigma(v) = \min \left\{ w(\text{cut}(S)) : v \in S \subseteq V, \perp \notin S \right\}, \quad (2)$$

where $\text{cut}(S) = \{uv \in E : u \in S, v \notin S\}$ and $w(\text{cut}(S)) = \sum_{e \in \text{cut}(S)} w(e)$. A minimising set $S^*(v)$ is a *resting cut* of v : the cheapest boundary separating v from the medium.

Theorem 3.2 (Separation). *For every $v \in V \setminus \{\perp\}$:*

- (i) $\sigma(v)$ is attained by at least one $S^*(v)$;
- (ii) $\sigma(v) \geq \beta > 0$;
- (iii) $\sigma(v)$ equals the value of a maximum v - $\perp$ flow, and is computable in strongly polynomial time.

Proof. (i) The feasible family $\{S : v \in S, \perp \notin S\}$ is a nonempty subset of the finite power set 2^V (it contains $\{v\}$), and $w(\text{cut}(\cdot))$ is a real-valued function on it, so a minimiser exists.

(ii) Let S be feasible. Since $\perp$ is adjacent to every other vertex ([Theorem 2.1](#)) and $v \in S$ while $\perp \notin S$, the edge $v\perp \in E$ has exactly one endpoint in S , hence $v\perp \in \text{cut}(S)$ and $\text{cut}(S) \neq \emptyset$. Therefore $w(\text{cut}(S)) \geq w(v\perp) \geq \beta$ by [Axiom 1](#). Taking the minimum over feasible S preserves the bound.

(iii) The right-hand side of (2) is by definition the minimum weight of a v - $\perp$ cut in G with capacities w . By the max-flow min-cut theorem ([Ford and Fulkerson, 1956](#)) this equals the maximum v - $\perp$ flow value, and it is computed in strongly polynomial time by, for instance, the Edmonds–Karp procedure ([Edmonds and Karp, 1972](#)). $\square$

Remark 3.3 (What (ii) rules out). Part (ii) is the reason a psychon has somewhere to be. Without it a vertex could be separated from the medium at cost approaching zero, and the statement “the agent is at v ” would carry no information: the agent would be at v and, to within any tolerance, equally at $\perp$. The floor makes occupancy quantal. It is also what makes the record of [Axiom 3](#) meaningful, since a step that crossed no boundary of positive weight would be a step between two things that were not two.

Proposition 3.4 (Resting cuts need not be unique, and this matters). *There exist agent graphs and vertices v with two distinct minimisers $S_1^* \neq S_2^*$ of (2).*

Proof. Take $V = \{\perp, v, a, b\}$ with $ab \notin E$ and all remaining edges present, weighted $w(v\perp) = 3$, $w(a\perp) = w(b\perp) = 1$, $w(va) = w(vb) = 1$; every weight is at least $\beta = 1$, so [Axiom 1](#) holds. Then

$$\begin{aligned} w(\text{cut}(\{v\})) &= w(v\perp) + w(va) + w(vb) = 3 + 1 + 1 = 5, \\ w(\text{cut}(\{v, a\})) &= w(v\perp) + w(vb) + w(a\perp) = 3 + 1 + 1 = 5, \\ w(\text{cut}(\{v, b\})) &= w(v\perp) + w(va) + w(b\perp) = 3 + 1 + 1 = 5, \\ w(\text{cut}(\{v, a, b\})) &= w(v\perp) + w(a\perp) + w(b\perp) = 3 + 1 + 1 = 5, \end{aligned}$$

and these are all the feasible sets. So all four are minimisers, and $\sigma(v) = 5$ is attained non-uniquely. $\square$

Non-uniqueness is not a defect to be broken by a tie rule. It is the finite shadow of the fact that “what counts as this thing rather than its surroundings” can be genuinely underdetermined by the graph, and any implementation that silently canonicalises the choice is asserting a determination the graph does not make. We return to this in [Invariant 5](#).

4 Recognition is search

Definition 4.1 (Retrieval signature). An operation ρ on the agent has the *retrieval signature* if there exist vertices u, v with $uv \notin E$ such that ρ carries the agent from u to v , and ρ leaves the record m unchanged.

Retrieval, in the ordinary sense, is exactly this: going straight to the stored item, at a cost independent of where one was. The next result says the runtime has no such operation, and that this follows twice over.

Theorem 4.2 (Recognition–search identity). *No operation of the runtime has the retrieval signature. Consequently, for every arrival at a vertex there is a walk realising it, and the record strictly increases along that walk.*

Proof. Suppose ρ has the retrieval signature, carrying the agent from u to v with $uv \notin E$.

First, ρ violates [Axiom 2](#), which permits arrival only at a vertex adjacent to the current one; since $uv \notin E$, v is not adjacent to u .

Second, and independently of [Axiom 2](#): by [Axiom 3](#) the record is monotone and every committed step increments it by at least $\varphi > 0$, so an operation that changes the occupied vertex while leaving m unchanged commits no step. But then the agent occupies v having committed nothing, and by monotonicity of m the runtime cannot distinguish the state after ρ from the state before it. Two distinct occupancies with an identical record are, by [Theorem 5.2](#) below, the same emitted state, so ρ is not an operation of the runtime at all: it has no effect the runtime can represent.

Hence no such ρ exists. Every arrival is therefore an adjacency step ([Axiom 2](#)), a finite composition of which is a walk, and each step increments m by at least φ . $\square$

Corollary 4.3 (Memory and inference are not distinguishable in the runtime). *There is no predicate on runtime states that holds of arrivals-by-recall and fails of arrivals-by-derivation.*

Proof. By [Theorem 4.2](#) both are walks on G satisfying [Axioms 2](#) and [3](#). A predicate separating them would have to be a function of the walk beyond its status as a walk; the only such data are the identities of intermediate vertices, and by [Theorem 5.3](#) below these are not present in any emitted state. A predicate on runtime states is by definition a function of emitted state. $\square$

Remark 4.4. [Theorem 4.3](#) is not scepticism about the distinction. It is a statement about this instrument: an agent that answers “I remember” and an agent that answers “I worked it out” have performed the same kind of operation, and the runtime is not entitled to label one of them. The second argument in the proof of [Theorem 4.2](#) is worth isolating, because it does not use [Axiom 2](#) at all: even an implementation that permitted jumps would find retrieval incoherent, since under a monotone record a costless jump is invisible.

5 The psychon and path opacity

Definition 5.1 (Psychon). A *psychon* on an agent graph G is a pair $\Psi = (\gamma, v)$ where $\gamma = (v_0, v_1, \dots, v_k)$ is a walk in G (so $v_i v_{i+1} \in E$ for all $i < k$) and $v = v_k$. Its *record* is $m(\Psi) = \sum_{i < k} c(v_i v_{i+1})$ where $c(e) \geq \varphi$ is the committed cost of traversing e , per [Axiom 3](#).

Definition 5.2 (Emitted state). The *emitted state* of a psychon is the pair

$$\pi(\Psi) = (v, m(\Psi)) \in V \times \mathbb{R}_{\geq 0}. \quad (3)$$

An agent acts only through π : no operation of the runtime takes as input, or produces as output, any function of γ not factoring through π .

Theorem 5.3 (Path opacity). *Let G be an agent graph with $n \geq 3$. Then there exist psychons $\Psi_1 = (\gamma_1, v)$ and $\Psi_2 = (\gamma_2, v)$ with $\gamma_1 \neq \gamma_2$, the same terminus v , and $m(\Psi_1) = m(\Psi_2)$. Hence π is not injective, and the interior of a psychon is not a function of its emitted state.*

Proof. Since $n \geq 3$ there are distinct $a, b \in V \setminus \{\perp\}$. The medium is adjacent to all vertices, so $\perp a, \perp b \in E$. Consider

$$\gamma_1 = (\perp, a, \perp, b, \perp), \quad \gamma_2 = (\perp, b, \perp, a, \perp), \quad (4)$$

both walks in G with the same terminus $v = \perp$. They are distinct as sequences, since $a \neq b$ gives $\gamma_1(1) = a \neq b = \gamma_2(1)$.

Both traverse the same multiset of edges, namely $\perp a$ twice and $\perp b$ twice, differing only in the order. Since m is by [Theorem 5.1](#) the sum of committed costs over traversed edges, and addition is commutative,

$$m(\Psi_1) = 2c(\perp a) + 2c(\perp b) = m(\Psi_2), \quad (5)$$

for any cost function c . Hence $\pi(\Psi_1) = (\perp, m(\Psi_1)) = (\perp, m(\Psi_2)) = \pi(\Psi_2)$ while $\Psi_1 \neq \Psi_2$, so π is not injective. $\square$

Corollary 5.4 (No self-certification). *An agent cannot emit a warrant that its own terminus was reached by one walk rather than another.*

Proof. Any such warrant is an emission, hence by [Theorem 5.2](#) a function of $\pi(\Psi)$. By [Theorem 5.3](#) there are $\Psi_1 \neq \Psi_2$ with $\pi(\Psi_1) = \pi(\Psi_2)$; a function of π takes the same value on both, so it does not distinguish them. $\square$

Corollary 5.5 (Individuation). *Two agents pruned to the same graph, having emitted the same (v, m) , may nonetheless be distinct psychons; and once $m > 0$ the pair (γ, v) is not reproducible from the pruning record.*

Proof. Immediate from [Theorem 5.3](#) for the first claim. For the second: the pruning record determines G and hence the set of available walks, but by [Axiom 3](#) the record is monotone and by [Axiom 4](#) no walk is forced, so the particular walk taken is not a function of G . $\square$

This is the promised completion of the parent construction. Before contact, pruning is deterministic and two agents from the same record are the same object; after the first committed step, they are psychons with different interiors, and no emission recovers the difference.

6 Receiver relativity and drift

So far one agent. We now put two or more on the same event and ask what they end up holding.

Definition 6.1 (Registration). Let $A_1, \dots, A_N$ be agents with graphs $G_1, \dots, G_N$, and let x be an event. A *registration* of x in A_i is a vertex $v_i \in V_i$ at which a walk prompted by x terminates, together with its resting cut $S^*(v_i) \subseteq V_i$ (Theorem 3.1).

Theorem 6.2 (Receiver relativity). *Let x be a single event received by agents A_1, A_2 with $G_1 \neq G_2$. Then:*

- (i) *the registrations $S^*(v_1) \subseteq V_1$ and $S^*(v_2) \subseteq V_2$ are subsets of different vertex sets and need not correspond under any graph morphism;*
- (ii) *each is a correct minimiser of (2) on its own graph;*
- (iii) *there is no graph on which both are cuts, hence no third party relative to which one is nearer to x than the other.*

Proof. (i) By construction $S^*(v_i)$ is defined as a subset of V_i . Since $G_1 \neq G_2$ there need be no bijection $V_1 \rightarrow V_2$ preserving w ; pruning against distinct demographic records produces distinct induced subgraphs, and Theorem 3.4 shows minimisers are not even unique within one graph.

(ii) Correctness is a property internal to (2): $S^*(v_i)$ minimises $w(\text{cut}(\cdot))$ over the feasible family of G_i . That is a finite optimisation with an attained minimum by Theorem 3.2(i), and it makes no reference to G_{3-i} .

(iii) Suppose H were a graph on which both $S^*(v_1)$ and $S^*(v_2)$ are cuts, so that their weights are comparable as separations in H . Then H contains $V_1 \cup V_2$ and assigns weights to edges within each; but the weight of $\text{cut}(S^*(v_1))$ computed in H includes edges to $V_2 \setminus V_1$, which are absent from G_1 and whose weights are not determined by either agent graph. So the comparison requires data not contained in G_1 , G_2 , or the event; by Axiom 4 no agent holds it, and by Theorem 5.2 no emission supplies it. $\square$

Corollary 6.3 (Divergence is not error). *If A_1 and A_2 register the same event differently, no predicate of the runtime marks either registration as incorrect.*

Proof. By Theorem 6.2(ii) both are correct on their own graphs, and by (iii) there is no graph relative to which they can be jointly scored. A predicate marking one incorrect would be a function of data the runtime does not hold. $\square$

Remark 6.4 (Falsity is an exterior relation). Theorem 6.3 does not say nothing is ever false. It says falsity is not a unary predicate on an agent state. To call a registration false is to compare a pair—the agent and something else—and the something else is always another graph, held by another agent, with the same standing. An observer may of course fix a reference graph and grade everything against it; that observer is then a further agent, and Theorem 6.2 applies to them too. What the runtime must not do is privilege one such graph internally, which is the content of Invariant 6.

Definition 6.5 (Reshaping and drift). A *reshaping* of an agent graph is a map $g : G \mapsto G'$ altering weights, arising from a registration. For a chain of receivers $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_n$, each passing on what it registered, the *composed drift* is

$$G_n = (g_n \circ g_{n-1} \circ \dots \circ g_1)(G_0). \quad (6)$$

Theorem 6.6 (Drift has no author). *Let $n \geq 2$ and suppose each g_i is non-trivial, i.e. $g_i \neq \text{id}$. Then in general $G_n \neq g_i(G_0)$ for every i , and no g_i is distinguished by any function of the emitted states $\{\pi_i\}_{i=1}^n$.*

Proof. The first claim is immediate: composition of non-identity weight maps differs from each factor unless the others cancel, which is non-generic.

For the second, suppose F is a function of $\{\pi_i\}$ selecting some index j as the author. Each $\pi_i = (v_i, m_i)$ by [Theorem 5.2](#). Consider the chain in which A_j and A_{j+1} exchange their reshaping, $g'_j = g_{j+1}$ and $g'_{j+1} = g_j$. Because composition of weight maps on disjoint edge supports commutes, the composite G_n is unchanged; and because [Theorem 5.3](#) makes π_i independent of which walk produced it, the emitted states can be arranged identical under the exchange. Then F returns j in both, but the reshaping performed by A_j differs between them. So F does not select the agent that performed a given reshaping. $\square$

Corollary 6.7 (Rumour). *A registration that has passed through $n \geq 2$ receivers is a composed drift: it is held by all of them, originated by none identifiably, and is not false in the sense of [Theorem 6.3](#).*

This is the formal content of the phenomenon the instrument is built to observe. A proposal circulates, everyone who holds it holds something slightly different, each holding is correct on its own graph, and no participant is the source. The parent manuscript proves the runtime cannot attribute; [Theorem 6.6](#) says there is nothing to attribute.

7 Acts and the direction of the gap

We can now answer question (c) of the introduction. The difficulty is that the obvious taxonomies are unavailable: by [Theorem 5.4](#) an agent cannot certify its own interior, so intent is not emitted; by [Theorem 6.3](#) the runtime cannot evaluate a content; and by [Theorem 5.3](#) the walk is not visible. What remains is the effect an act has, and it turns out that is enough—and that it yields exactly two kinds.

Definition 7.1 (Residual gap). For an agent A engaged with a matter, the *residual gap* $D(A) \geq 0$ is the total weight of the resting cut at its current terminus that remains untraversed:

$$D(A) = w(\text{cut}(S^*(v))) - \rho(A), \quad (7)$$

where $\rho(A)$ is the weight already committed against that cut.

By [Theorem 3.2\(ii\)](#), $w(\text{cut}(S^*(v))) \geq \beta > 0$, so the gap starts positive; by [Axiom 4](#) it is never exhausted by any finite walk.

Theorem 7.2 (Exchange does not close). *Let two agents exchange acts about a matter. While $D > 0$ on either side, a further act is available, and no state of the exchange is a terminal state in which both gaps are zero.*

Proof. Suppose $D(A_1) = D(A_2) = 0$. Then each agent has committed weight equal to the full resting cut at its terminus, so each has traversed every edge of $\text{cut}(S^*(v_i))$. By [Axiom 4](#) there remains in G_i a vertex not on the walk and an edge not traversed; take such an edge e incident to $S^*(v_i)$ if one exists. If none exists, $S^*(v_i)$ is a connected component separated from the rest, and since $\perp \notin S^*(v_i)$ while $\perp$ is adjacent to every vertex ([Theorem 2.1](#)), some edge from $S^*(v_i)$ to $\perp$ is present and by [Axiom 1](#) has weight at least $\beta > 0$; traversing it is an available act that returns the gap to a positive value. Either way an act is available, contradicting terminality. $\square$

Remark 7.3. [Theorem 7.2](#) is why the instrument does not terminate on agreement. Exchange ends by exhaustion of what the participants are willing to spend, not by the matter closing, because the matter does not close. This is the non-convergence of the parent title, localised to a pair.

Definition 7.4 (Report and question). Let A act toward A' , and write D, D' for the gaps immediately before and $\widehat{D}, \widehat{D}'$ immediately after. The act is a

- *report*, written r , if $\widehat{D} < D$: it lowers the gap on the acting side;
- *question*, written q , if $\widehat{D}' > D'$: it raises the gap on the receiving side.

Theorem 7.5 (Act direction). *Theorem 7.4 is well posed and exhaustive:*

- (i) *every act with any effect is a report, a question, or both;*
- (ii) *the two are independent, i.e. each of the four combinations of $\{is/is\ not\ a\ report\} \times \{is/is\ not\ a\ question\}$ is realisable except the doubly-negative one, which has no effect;*
- (iii) *the classification is decidable from emitted states alone, without reference to content, intent, or interior.*

Proof. (iii) first, since it constrains the rest. By Theorem 7.1, D is a function of v and the committed weight ρ , both components of the emitted state π of Theorem 5.2 (the resting cut is computed from v and G by Theorem 3.2(iii)). So the sign of $\widehat{D} - D$ is computable from π before and after, which is what the runtime holds.

(i) Let α be an act with some effect. By Axiom 3 it commits at least $\varphi > 0$, so it changes ρ on at least one side. If it changes $\rho(A)$ upward, then by Theorem 7.1 $\widehat{D} < D$ and α is a report. If it changes the receiving side, it does so by prompting a walk in A' (Axiom 2), which by Theorem 4.2 moves v' ; a move of terminus changes $S^*(v')$ and hence $w(\text{cut}(S^*(v')))$. If that weight rises, $\widehat{D}' > D'$ and α is a question. If it does not rise, the receiving side has committed against its cut without the cut growing, which is a report by A' rather than an effect of α . So an act with an effect falls in at least one class.

(ii) Realisability. A report that is not a question: A traverses an edge of its own resting cut and A' takes no step, so D falls and D' is unchanged. A question that is not a report: A acts so as to move v' to a vertex of larger separation cost while committing nothing against its own cut, possible since by Theorem 3.4 distinct vertices of G' have distinct σ values in general. Both at once: an act doing each of the above. Neither: an act committing nothing, excluded by Axiom 3 from having an effect. $\square$

Corollary 7.6 (A question is not a request for content). *The predicate “ α is a question” is invariant under relabelling of what α says.*

Proof. By Theorem 7.5(iii) the predicate is a function of emitted states, which by Theorem 5.2 contain (v, m) and no content. $\square$

Remark 7.7 (Why this is the definition worth having). An agent that asks in order to learn and an agent that asks in order to unsettle perform the same act under Theorem 7.4, and the instrument is right not to separate them: by Theorem 5.4 neither could certify which it was doing, including to itself. What the definition does capture is the asymmetry that matters for a deliberation: a report spends the speaker down, a question spends the hearer up. A population of agents that only report converges to silence by exhaustion; a population that also questions does not, and Theorem 7.2 says it cannot. This is the mechanism by which the agents of the instrument ask questions *for their own graph* rather than on a script: raising D' is the only act whose cost is paid by someone else.

8 Adoption without shared interiors

The instrument must be able to represent a practice spreading: one agent comes to do what another does. Theorem 5.3 and Theorem 6.2 between them forbid this happening by transfer of interior state. The following says it happens anyway, through boundaries.

Definition 8.1 (Boundary condition). The *boundary condition* of agent A at terminus v is the weighted edge set $\partial(A) = \text{cut}(S^*(v))$ together with the map $e \mapsto w(e)$ on it. Two agents *match* at tolerance ε if their boundary conditions agree, as weighted multisets, to within ε .

Theorem 8.2 (Behavioural adoption). *Let A_1, A_2 have graphs $G_1 \neq G_2$ and match at tolerance ε . Then for every act α classified by Theorem 7.4, the classification of α agrees between A_1 and A_2 up to $O(\varepsilon)$. In particular, matching boundaries induce matching behaviour without any correspondence between V_1 and V_2 .*

Proof. By Theorem 7.5(iii) the classification of α is the sign of a difference of gaps, and by Theorem 7.1 the gap is $w(\text{cut}(S^*(v))) - \rho$. The first term is a function of the boundary condition alone (Theorem 8.1); the second is the committed record, which is emitted. If the boundary conditions agree to within ε as weighted multisets, the first terms agree to within ε , so the gaps agree to within ε at equal ρ , and the signs of their differences agree wherever those differences exceed ε in magnitude. By Axiom 3 every act moves ρ by at least φ , so for $\varepsilon < \varphi$ the classification agrees exactly. No step of this argument refers to V_1 , V_2 , or any map between them. $\square$

Corollary 8.3 (Adoption is not inspection). A_2 may come to behave as A_1 does without any emission of A_1 having determined any interior of A_2 .

Proof. By Theorem 8.2 matching $\partial(A_1)$ and $\partial(A_2)$ suffices for matching behaviour, and ∂ is a function of the resting cut, which by Theorem 5.3 does not determine the walk. Distinct interiors are compatible with identical boundaries. $\square$

Proposition 8.4 (Threshold). *Let a population of N agents be coupled with strength K and let Δ measure the spread of their boundary conditions. Matching at tolerance ε is attained for all pairs only if $K > K_c$ with $K_c = \Theta(\Delta)$; below K_c the unmatched configuration is stable.*

Proof sketch. This is the coordination theorem of the parent manuscript applied to ∂ rather than to internal state, and the argument is unchanged: coupling must overcome heterogeneity, the critical value scales linearly in the spread, and below it the incoherent configuration is linearly stable. We do not reproduce it; we note only that the quantity being locked here is a boundary, which is exactly the quantity Theorem 8.2 shows is sufficient. $\square$

Remark 8.5 (Erosion under replacement). Adoption is not free. Replacing a part of an agent graph is a removal followed by an addition, and each leaves a residue: the removal discards the weights of edges between the removed part and the remainder, and the addition must fix weights not determined by either. By Axiom 3 these are committed operations, so the record grows monotonically under replacement and never returns. After n replacements the accumulated residue is $\Theta(n)$ under statistically similar replacements. An agent that has adopted much is therefore not the agent that began, and by Theorem 5.5 there is no operation restoring it. We record this because it is the honest cost of Theorem 8.2: a population that converges behaviourally has paid for it in individuation, and the instrument should display that rather than hide it.

9 The continuous correspondence

The discrete account above is complete as it stands. This section adds a continuous description of the *attracting set* of an agent, because the parent manuscript proves such an attractor exists without constructing one, and because an explicit potential is useful for implementation. Everything here is defined on one agent graph. Section 9.1 states what this does *not* license.

Definition 9.1 (Interior coordinates). Fix an agent A with graph G . Its *interior coordinates* are

$$q = (R, \sigma^2, s) \in \mathcal{M} = [0, 1] \times [\sigma_{\min}^2, \sigma_{\max}^2] \times [0, 1]^d, \quad (8)$$

where R is the coherence of the agent walk (the fraction of recent steps lying inside the current resting cut), σ^2 is the dispersion of its terminus over recent steps, and s collects d bounded summaries of the emitted record. $\mathcal{M}$ is compact and convex.

Definition 9.2 (Drive potential). A *drive potential* for $\mathbf{A}$ is a function $\Phi : \mathcal{M} \rightarrow \mathbb{R}$ of the form

$$\Phi(q) = \underbrace{\frac{1}{2}(K_c - K)R^2 + \frac{1}{4}KR^4}_{\text{coherence}} + \underbrace{\theta\sigma^2 + \frac{\theta}{K\sigma^2}}_{\text{dispersion floor}} - \underbrace{aRe^{-\sigma^2/\tau}}_{\text{coupling}} + \underbrace{V(s)}_{\text{boundary}}, \quad (9)$$

with $K, \theta, a, \tau > 0$ and V smooth, bounded and convex on $[0, 1]^d$.

Theorem 9.3 (Attractor). Suppose Φ is λ -strongly convex on $\mathcal{M}$ for some $\lambda > 0$. Then the gradient flow

$$\dot{q} = -\nabla\Phi(q) \quad (10)$$

has a unique fixed point $q^* \in \mathcal{M}$, and every trajectory converges to it exponentially:

$$\|q(t) - q^*\| \leq e^{-\lambda t} \|q(0) - q^*\|. \quad (11)$$

Moreover the discretisation $q_{k+1} = q_k - \delta \nabla\Phi(q_k)$ satisfies $\|q_k - q^*\| \leq (1 - \lambda\delta)^k \|q_0 - q^*\|$ for $\delta \leq 1/L$, where L is the Lipschitz constant of $\nabla\Phi$.

Proof. $\mathcal{M}$ is compact and convex and Φ is λ -strongly convex on it, so Φ attains its minimum at a unique q^* , which is the unique stationary point and hence the unique fixed point of (10). Let $W(t) = \frac{1}{2}\|q(t) - q^*\|^2$. Then $\dot{W} = -\langle q - q^*, \nabla\Phi(q) \rangle \leq -\lambda\|q - q^*\|^2 = -2\lambda W$ by strong convexity together with $\nabla\Phi(q^*) = 0$, and Grönwall gives the stated bound. The discrete estimate is the standard contraction bound for gradient descent on a λ -strongly convex, L -smooth objective with step $\delta \leq 1/L$ (Boyd and Vandenberghe, 2004). $\square$

Proposition 9.4 (Agreement with the discrete account). The step bound $\delta \leq 1/L$ of Theorem 9.3 is compatible with Axiom 3: the act floor φ bounds δ from below, so the admissible range is $\varphi \leq \delta \leq 1/L$, nonempty whenever $L \leq 1/\varphi$. Within it, the fixed point q^* is the continuous image of the set of termini at which the residual gap is stationary.

Proof. By Axiom 3 no committed step is smaller than φ , so a discretisation of (10) realisable by the runtime has $\delta \geq \varphi$. Stationarity of Φ means $\nabla\Phi(q^*) = 0$; by Theorem 9.1 the components of q are functions of the terminus and the record, so a stationary q is one at which further committed steps do not change the coherence or dispersion of the walk, which by Theorem 7.1 is the condition that D is stationary. $\square$

Remark 9.5 (What the potential adds and what it does not). Theorem 9.3 supplies a constructive Φ for the attractor the parent manuscript proves abstractly. It does not weaken Theorem 7.2: q^* is a stationary point of the coherence of a walk, not a state of zero gap. By Axiom 4 the gap remains positive at q^* , so an agent at its attractor is still acting. The attractor is a habit, not a conclusion.

9.1 What is not an observable

Proposition 9.6 (No global order parameter). Let R_i be the coherence of agent $\mathbf{A}_i$ per Theorem 9.1. Then no aggregate $R_{\text{pop}} = F(R_1, \dots, R_N)$ is an observable of the runtime.

Proof. Each R_i is defined relative to the resting cut of $\mathbf{A}_i$ on G_i (Theorem 9.1). By Theorem 6.2(iii) there is no graph on which the resting cuts of two distinct agents are jointly cuts, so the R_i are not commensurable quantities and F is not well defined on them. Even granting a fixed normalisation, F would be a function of the interiors of all agents; by Theorem 5.3 interiors are not emitted, so F is not computable from what the runtime holds. $\square$

Corollary 9.7 (The instrument still reports nothing it should not). *Adjoining Theorem 9.2 to the runtime introduces no verdict, no progress measure, and no attribution.*

Proof. A verdict or progress measure over the population is an aggregate of the kind excluded by Theorem 9.6. A per-agent Φ is not one: it is defined on G_i alone, is receiver-relative by Theorem 6.2, and by Theorem 9.5 its minimum is not a terminal state. Attribution is excluded by Theorem 6.6. $\square$

Remark 9.8 (On the temptation). The result above is the whole reason this section is written as carefully as it is. A potential and an order parameter are exactly the objects one reaches for when asked how the population is doing, and computing R_{pop} would be a few lines. Theorem 9.6 says the resulting number would not mean anything: not that it would be imprecise, but that the quantities averaged are defined on incommensurable graphs. The instrument may compute Φ per agent because each such Φ is that agent view; it may not compute the population figure, because there is no agent whose view it would be.

10 Implementation invariants

Each invariant is a decidable predicate on the runtime, paired with the result it enforces.

Invariant 1 (Floor). Every edge weight in every pruned graph is at least $\beta > 0$, checked at construction. Enforces Axiom 1 and hence Theorem 3.2(ii).

Invariant 2 (Monotone record). The record m of every agent is non-decreasing across every operation. No interface decrements it. Enforces Axiom 3 and Theorem 4.2.

Invariant 3 (No retrieval). No operation sets an agent terminus to a vertex non-adjacent to its current terminus. Enforces Axiom 2; violation would reintroduce the retrieval signature of Theorem 4.1.

Invariant 4 (Emission is the reduced pair). Every value crossing the agent boundary factors through $\pi(\Psi) = (v, m)$. No serialisation exposes γ . Enforces Theorem 5.3 and Theorem 5.4.

Invariant 5 (Explicit non-uniqueness). Where (2) has multiple minimisers, the implementation records that fact rather than silently selecting one; any selection is marked as a choice of the implementation and not a determination of the graph. Enforces the reading of Theorem 3.4.

Invariant 6 (No reference graph). No agent graph is designated as the standard against which others are scored, and no code path compares registrations across agents for correctness. Enforces Theorem 6.2 and Theorem 6.3.

Invariant 7 (No authorship field). No representation of a circulating registration carries an originator, and none is reconstructible from stored state. Enforces Theorem 6.6.

Invariant 8 (Acts classified by direction only). The report/question classification is computed from gaps before and after, never from content or a declared intent field. Enforces Theorem 7.5(iii) and Theorem 7.6.

Invariant 9 (No population aggregate). No route, log line, or stored value aggregates a per-agent coherence, potential, or gap across agents. Enforces Theorem 9.6 and Theorem 9.7.

11 Limitations

The continuous section is optional. Section 9 is a convenience for implementation, not a load-bearing part of the argument. Deleting it leaves Theorems 3.2, 4.2, 5.3, 6.2, 6.6, 7.5 and 8.2 intact. We flag this because the continuous apparatus is the part most likely to be over-read.

The threshold is imported. [Theorem 8.4](#) is stated as a sketch because its proof is the coordination theorem of the parent manuscript. We have applied it to a new quantity—the boundary condition rather than internal state—and that application is ours; the underlying threshold result is not.

Tolerance in adoption is a parameter. [Theorem 8.2](#) gives exact agreement only for $\varepsilon < \varphi$. For coarser matching the classification agrees only up to $O(\varepsilon)$, and the instrument should expose ε rather than fix it.

Non-uniqueness is under-explored. [Theorem 3.4](#) establishes that resting cuts may fail to be unique and [Invariant 5](#) says what to do about it, but we do not characterise when non-uniqueness occurs, nor how often it does on graphs produced by pruning against the Zürich substrate. That is an empirical question this paper does not answer.

12 Conclusion

We took the graph the parent construction produces and asked what lives on it. The answer is a psychon: a walk together with the vertex it has reached, carrying a monotone record. The walk is the agent individuality and is inaccessible; the pair of terminus and record is everything anyone else gets; and those two facts are the same fact, since it is precisely the non-injectivity of the emission map that makes two identically-pruned agents into two agents.

From this the conversational mechanics follow without further assumption. Recalling is searching, because a costless jump under a monotone record is invisible. One event registers differently in every receiver, each correctly, with no graph on which the registrations compete. A registration passed along a chain becomes a drift with no author. And an act is a report or a question according to which side of the exchange its cost falls on—the only distinction that survives when content, intent and interior are all unavailable, and, we have argued, the one that was doing the work anyway.

Finally, practices spread. Agents whose boundaries match behave alike regardless of what is behind those boundaries, which is how something can propagate through a population that shares no interior and inspects nothing. It is not free: every adoption is a replacement, every replacement commits record, and the record does not run backwards. A population that comes to agree has spent individuation to do it. The instrument is built to show that transaction rather than to score its outcome, and [Section 10](#) is the list of places where we have had to say so in code.

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